

Gig Economy Earnings Analysis

Modeling Driver Profitability for App-Based Delivery Work

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1 Introduction

1.1 Motivation and Problem Statement

Popularity has risen for app-based gig delivery work (platforms like DoorDash, Uber Eats, and Instacart), which has grown into a primary or supplemental source of income for millions of workers across the United States. Despite the massive market, current and potential drivers have almost no structured and accurate way to evaluate whether gig driving is financially worth it for their specific metrics. Reports of earnings vary widely across online forums, journals, and driver surveys. Additionally, they rarely account for hidden variables including fuel consumption, vehicle wear, idle time, and the geographic and time-of-day variability that shapes what any individual session actually pays.

This project addresses that problem directly. The end goal is to build a realistic earnings model based on real data that estimates expected pay ranges, accounts for earnings variance, and utilizes key factors such as vehicle type, market, time of day, and driver efficiency. This will inform drivers whether sessions will be profitable or not. The purpose of this analysis is intended to back a driver ROI tool that takes the user's key factor inputs and returns a data-backed assessment of whether gig driving is worth it for that person.

1.2 What This Project Aims to Solve

The main problem this project aims to solve is the absence of transparent and structured earnings data for gig drivers. Specifically, it aims to:

- Estimate realistic hourly and per-session earnings ranges under varying conditions
- Quantify the cost of fuel on different vehicle classes
- Identify which locations and time slots are most associated with higher net pay
- Model earnings variability, risk, and enjoyment, not just average outcomes
- Provide a foundation for a future interactive ROI web tool

In the end, given that a person has chosen to drive, what does the data say about them, how to do it most profitably, and what could the risks be?

1.3 Data Sources and Dataset Overview

Since there are no public dataset that directly captures DoorDash driver earnings at a session level, this project uses a synthesized dataset constructed to reflect accurate and realistic gig economy earnings patterns. The synthesis was put together with various publicly available reports from drivers and driver surveys, gig economy research reports, and published pay structure documentation from delivery platforms.

The primary dataset used in this analysis (`doordash_earnings.csv`) contains 2,000 rows and 35 columns. Each row represents one synthesized driving session. Key data dimensions include:

- Geographic identifiers: `region`, `census_region`
- Temporal identifiers: `time_slot` (morning, lunch, dinner, late_night)
- Driver characteristics: `driver_experience_tier`, `vehicle_type`, `vehicle_class`

- Operational metrics: hours_worked, active_hours, active_ratio_pct, total_deliveries, deliveries_per_hour, avg_miles_per_delivery
- Pay components: doordash_pay_usd, gross_base_pay_usd, gross_peak_pay_usd, challenge_bonus_usd, gross_tips_usd
- Cost components: resolved_mpg, avg_gas_price, gas_cost_usd, uses_fuel
- Derived earnings: gross_earnings_usd, net_earnings_usd, net_hourly_rate

1.4 Key Assumptions About the Data

Because the dataset is synthesized rather than directly observed, several assumptions underpin its construction and must be held in mind throughout all interpretations:

- Pay rates, tip rates, and delivery frequencies were seeded from publicly available driver surveys and benchmarks, not from proprietary platform data. They represent realistic approximations, not ground truth.
- Fuel costs were calculated using vehicle-class-specific MPG estimates sourced from EPA and NHTSA standards, combined with regional gas price data. Hybrid and electric vehicle costs use static MPGe estimates.
- The synthesized data does not account for platform changes or individual driver strategies that would alter outcomes of income in the real world.
- The synthesized data was intentionally designed to simulate real-world variance, but the AI model determined the distribution of each session; there is no real-world variance from real data.
- All findings from this analysis are held conditional on these synthesis assumptions. The findings aim to replicate what people might see in the real world; however, they describe what the model predicts, not what any individual driver will actually experience.

2 Data Preparation

2.1 Cleaning Pipeline

The data preparation phase was conducted in a dedicated Jupyter notebook (01_data_cleaning.ipynb) following a structured pipeline designed for reproducibility. Although the synthesized dataset arrived in relatively clean condition compared to raw public datasets, several processing steps were applied to ensure analytical integrity.

Null and missing value checks were performed across all 35 columns. Columns with boolean types (`uses_fuel`, `has_peak_pay`, `has_challenge_bonus`) were verified to contain only binary values with no unexpected nulls. Numeric columns were inspected for out-of-range values, and the `resolved_mpg` column received special treatment: electric vehicles were assigned a static EPA MPGe equivalent, bicycles were assigned a fuel cost of zero, and motorcycles used an NHTSA-sourced estimate. This vehicle class crosswalk was one of the more involved steps in the pipeline, as the raw vehicle classifications required reconciliation before fuel cost calculations could be applied consistently.

Gas cost calculations were computed as a derived column by joining vehicle MPG against regional average gas prices, then multiplying by total miles driven. Net earnings were then derived by subtracting gas cost from gross earnings. All intermediate calculations were preserved as separate columns rather than overwriting source values, so the cost decomposition remains auditable at every step.

2.2 Feature Engineering and Column Decisions

Several columns were intentionally excluded from the modeling feature set to reduce redundancy. Traffic density was excluded as a standalone feature because urban classification variables serve as sufficient proxies.

The `active_ratio_pct` column, the fraction of online time spent actively on a delivery versus waiting, was retained as a key efficiency metric. This column represents one of the most actionable variables for a driver: idle time is within a driver's partial control through market selection and shift timing. The `deliveries_per_hour` column serves a similar role and was retained alongside it.

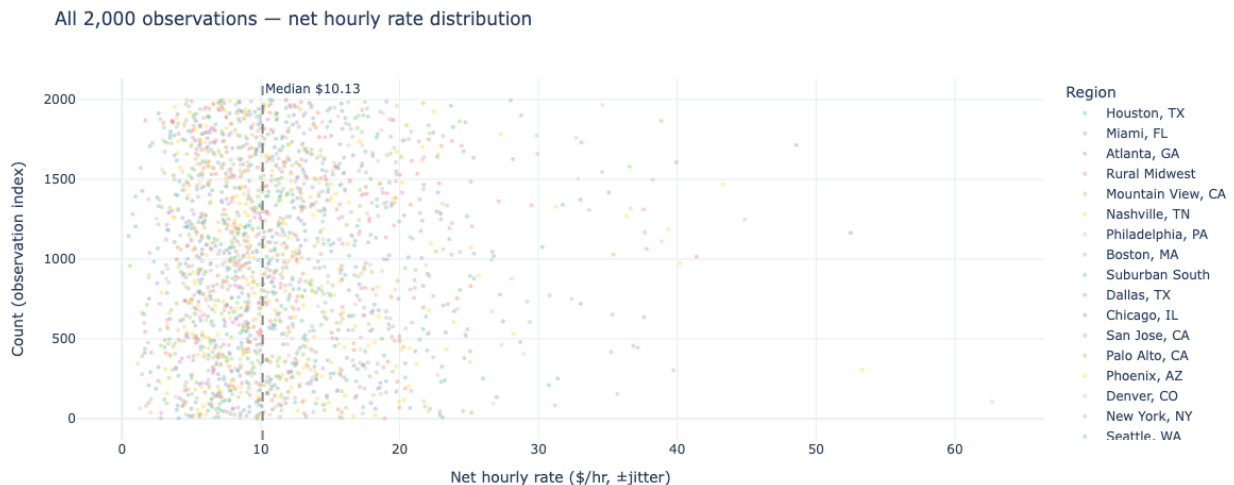
2.3 Dataset Schema

After cleaning, the final dataset used for all exploratory analysis contains 2,000 complete session records across 35 columns. The primary target variable for all downstream modeling is `net_hourly_rate`, which represents total earnings after fuel costs divided by total hours worked. This is the most honest single measure of driver profitability because it accounts for both the revenue and the most significant variable operating cost.

3 Exploratory Analysis and Chart Interpretations

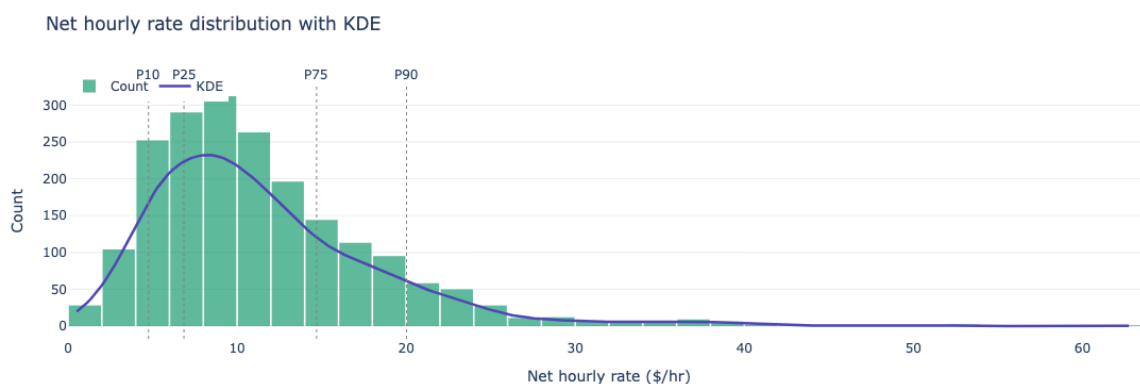
All charts in this section were generated using Plotly in a dedicated visualization notebook. Charts are interactive in the notebook environment. The interpretations below describe what each chart reveals and what practical conclusions can be drawn from it.

3.1 Scatter: All Observations



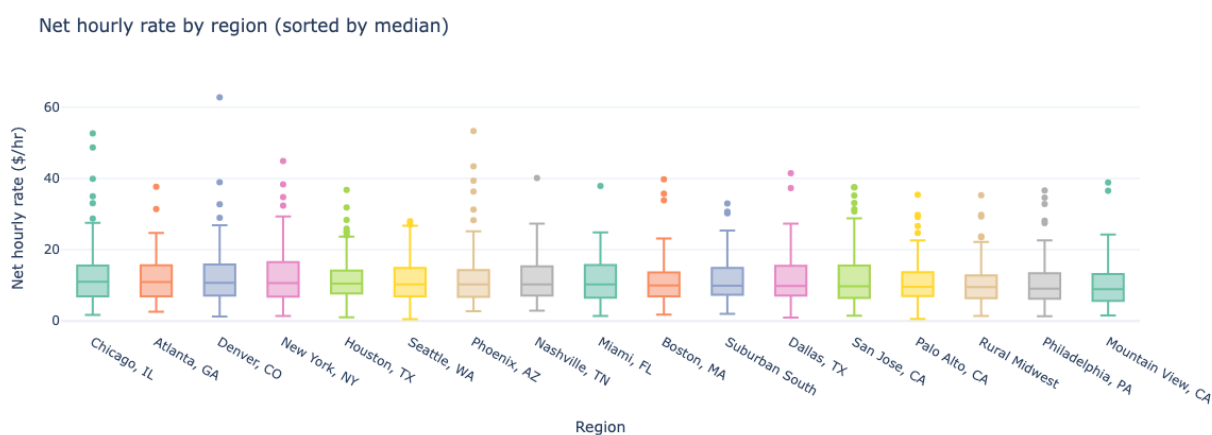
The first thing this chart tells you is that the overwhelming majority of sessions cluster tightly between \$0 and \$20 per hour, with the median landing at \$10.13. The density of points in that left band makes clear that a "typical" session is not particularly rewarding. What pulls the eye are the few but real observations stretching out to \$40, \$50, even \$60+ per hour. These aren't errors; they represent real, highly efficient sessions, but they're too rare for a driver to plan their finances around them. The regional coloring doesn't show dramatic separation, which is an indication that geography alone doesn't cleanly predict who earns more. The chart generated is most useful as a tool to help see and understand the data. This is the raw truth of what the data contains when looking at the net hourly rate.

3.2 Histogram and KDE Distribution



The distribution is right-skewed and unimodal, which is exactly what you'd expect from gig earnings data. The peak sits around \$7–10 per hour, meaning the most common session outcome is modest pay. The KDE curve confirms there's no secondary peak; this isn't a bimodal distribution with a "high earner" cluster hiding separately from a "low earner" cluster. It's one population with a long tail. The percentile markers are the most actionable part of this chart: P10 sits around \$3/hr, meaning the bottom 10% of sessions are barely covering costs, while P90 sits around \$20/hr, meaning genuinely strong sessions exist but represent only the top tenth of outcomes. For the ROI tool, the honest message is that a new driver should expect to land somewhere in that \$5–15 band far more often than anywhere else.

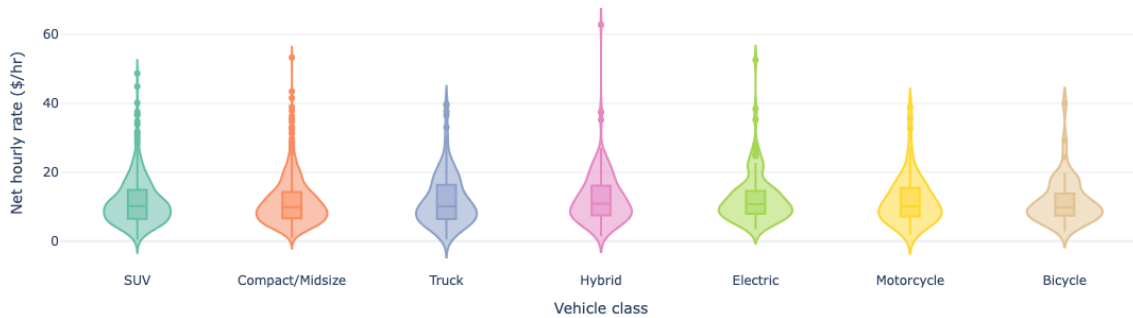
3.3 Box Plots by Region



The most striking finding here is how similar all the regions actually are. Chicago leads the sort order with the highest median, but the difference between first and last place is small, with every region sitting somewhere between roughly \$9 and \$13 per hour. What varies more noticeably is the spread. Denver and New York have the widest boxes and the most dramatic outliers, with individual sessions reaching \$60+, meaning those markets offer more upside but also more unpredictability. Mountain View and Philadelphia sit at the lower end with tighter boxes, suggesting more consistent but more modest earnings. The real takeaway for a driver is that market selection probably matters less than expected for the average session, but if you are chasing high-earning outlier sessions, larger and denser markets give you more opportunities to reach them.

3.4 Violin Plots by Vehicle Class

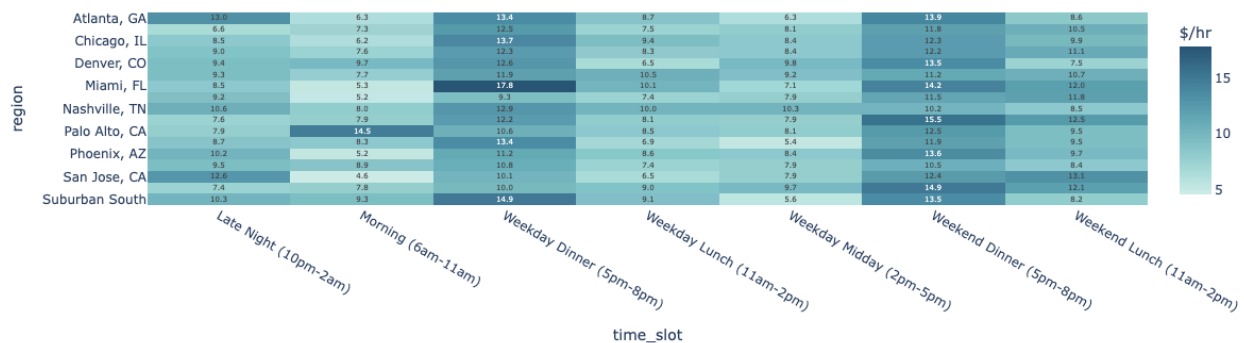
Earnings distribution by vehicle class



All seven vehicle classes share medians that are remarkably close together, clustered around \$8–13 per hour, so vehicle type does not dramatically change a typical session outcome. What does differ is the shape of the tails. Hybrid and Electric vehicles have the longest upward necks, meaning they occasionally produce sessions well above \$40/hr because their lower fuel cost per mile means more money stays in the driver's pocket on long, high-pay runs. SUVs and Compact/Midsize cars show fatter bodies at lower earnings levels, reflecting the drag of higher fuel consumption across a broad range of sessions. Bicycle stands out for being the narrowest violin overall, with almost no sessions above \$25/hr, which makes physical sense since limited speed and range cap both delivery volume and tip potential. Vehicle choice is ultimately more about managing your earnings ceiling and downside risk than it is about changing what a typical session pays.

3.5 Heatmap: Region × Time Slot

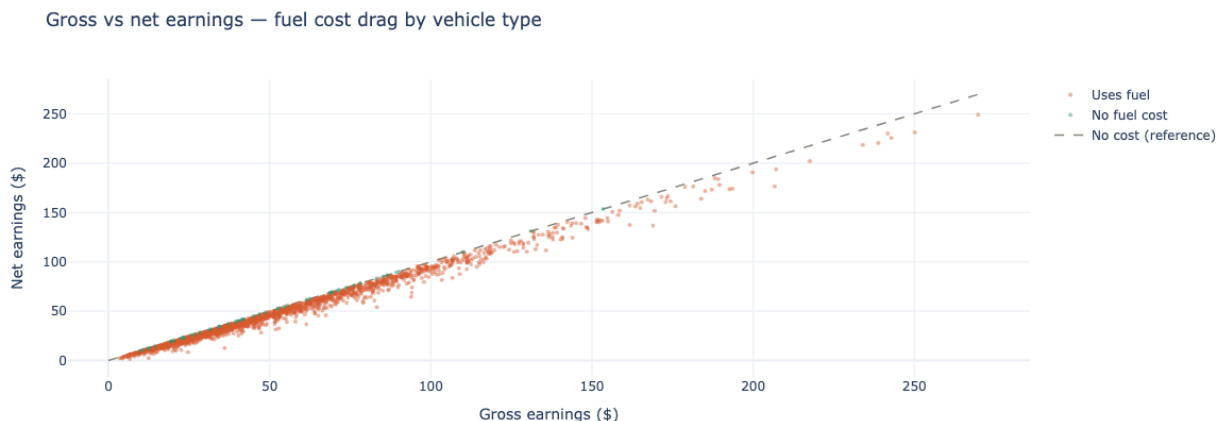
Median net hourly rate (\$/hr) — region × time slot



This is the most operationally useful chart in the suite. The highest-earning combinations are concentrated in specific pockets rather than spread evenly across the grid. Miami at Weekday Dinner stands out at \$17.80/hr, the single highest cell in the entire matrix. Palo Alto at Late Night hits \$14.50, and Weekend Dinner performs strongly across Atlanta, Denver, and Nashville. Morning slots are consistently lighter across nearly every region, confirming that early shifts underperform regardless of market. The practical implication is that time slot selection is one of the most actionable decisions a driver can make, and the best time to drive is not the same in

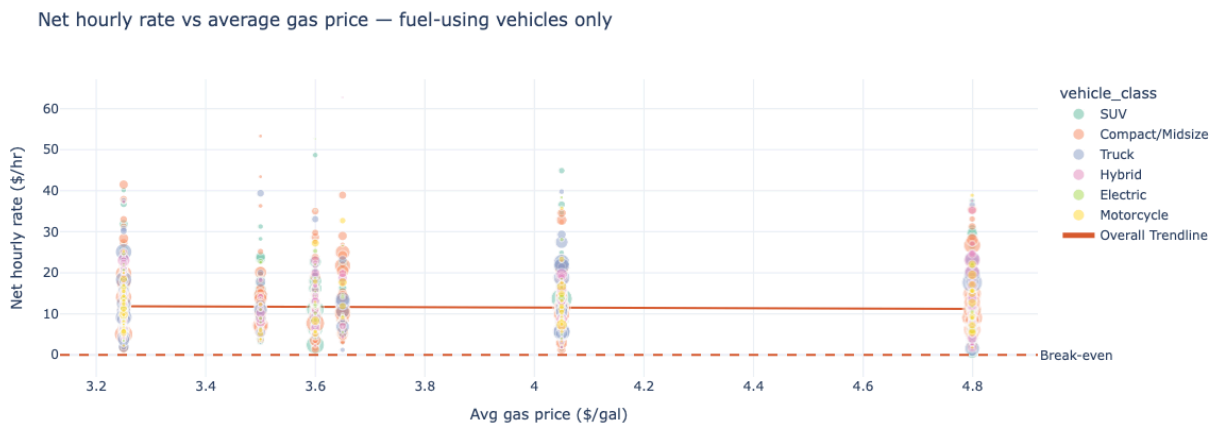
every city. This chart is essentially the scheduling recommendation engine for the ROI tool.

3.6 Gross vs Net Earnings Scatter



What immediately stands out is that the fuel and no-fuel populations are nearly on top of each other, both tracking closely along the 45-degree reference line. This tells you that at the session level, fuel cost is a real but not catastrophic drag on earnings. The gap between gross and net is present but modest for most sessions. Where separation becomes more visible is at higher gross earnings, where sessions above \$150 show more daylight between the two groups, meaning fuel costs compound as mileage increases. The chart also confirms that no vehicle class is producing sessions that fall dramatically below the reference line, which validates the cost calibration in the model. For the ROI tool, the message is that fuel matters most for drivers doing high-mileage sessions rather than short urban runs.

3.7 Gas Price vs Net Hourly Rate

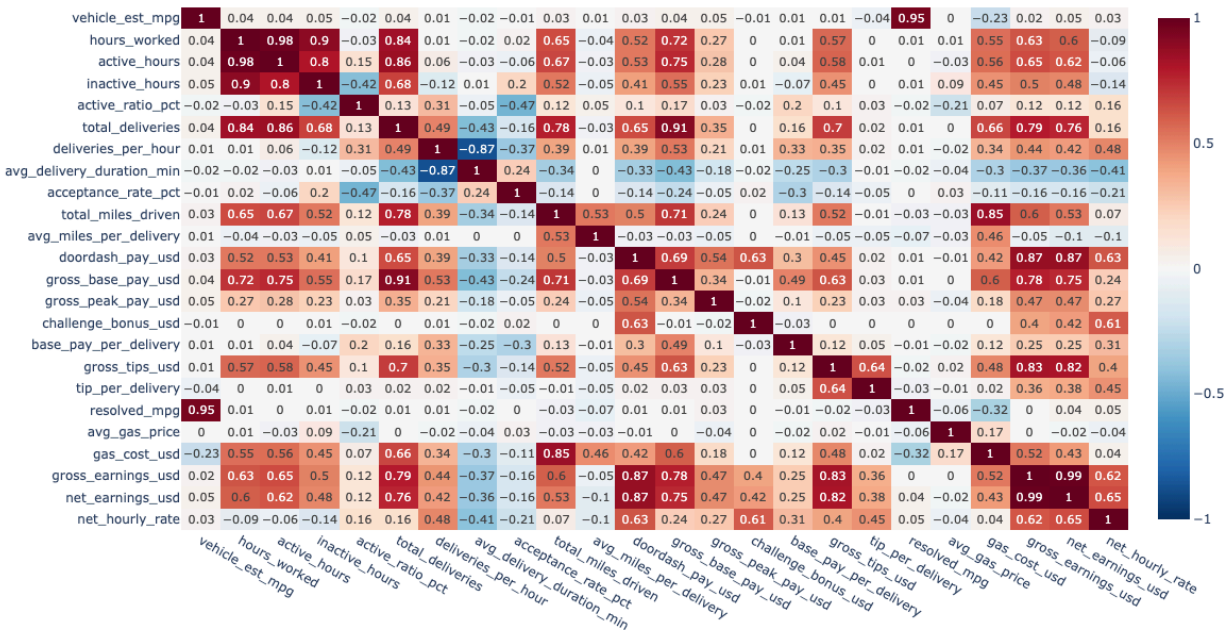


The trendline here is essentially flat, which is actually a meaningful finding rather than a boring one. Gas prices in the dataset cluster around four discrete bands and at none of those price points does the median net hourly rate shift dramatically. This suggests that within the gas price range present in the data, fuel costs are not the dominant driver of session-to-session earnings variance.

Delivery volume, tips, and active ratio appear to matter more. The wide vertical spread of points at every gas price level reinforces this, showing high-earning and low-earning sessions coexisting at both cheap and expensive gas prices. The break-even line confirms that even at \$4.80/gallon the model is not pushing drivers into negative territory, which is a useful validation of how fuel costs were calibrated.

3.8 Correlation Matrix

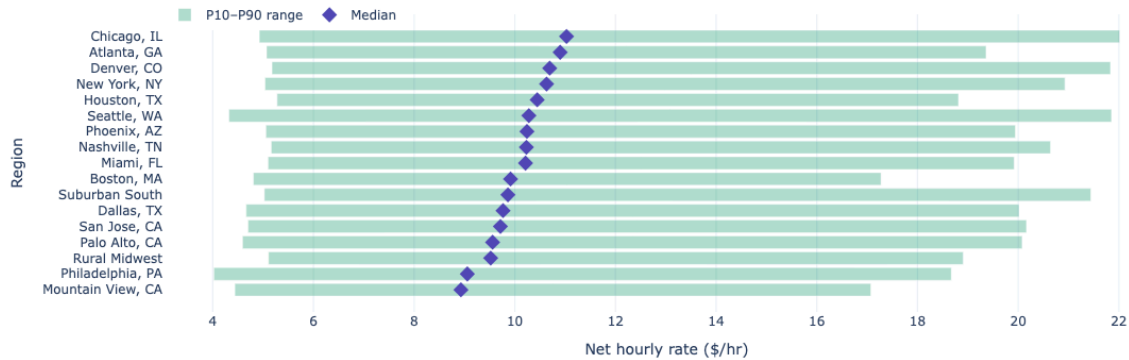
Pearson correlation matrix — all numeric features



The strongest correlations with net hourly rate come from gross earnings (0.65) and deliveries per hour (0.48). The 0.95 correlation between vehicle_est_mpg and resolved_mpg flags those as near-duplicate columns, and one should be dropped before any modeling begins. The moderate positive correlation between active ratio and net hourly rate (0.16) is modest but directionally consistent with what Chart 12 shows. The strong cluster of high correlations around hours worked, active hours, and total deliveries reflects heavy intercorrelation among time-based columns, and including all of them in a regression would create multicollinearity problems. The 0.61 between challenge bonus and peak pay is also worth noting, as those two features likely measure similar promotional pay structures and could reasonably be consolidated into a single bonus pay variable.

3.9 Quantile Bands by Region

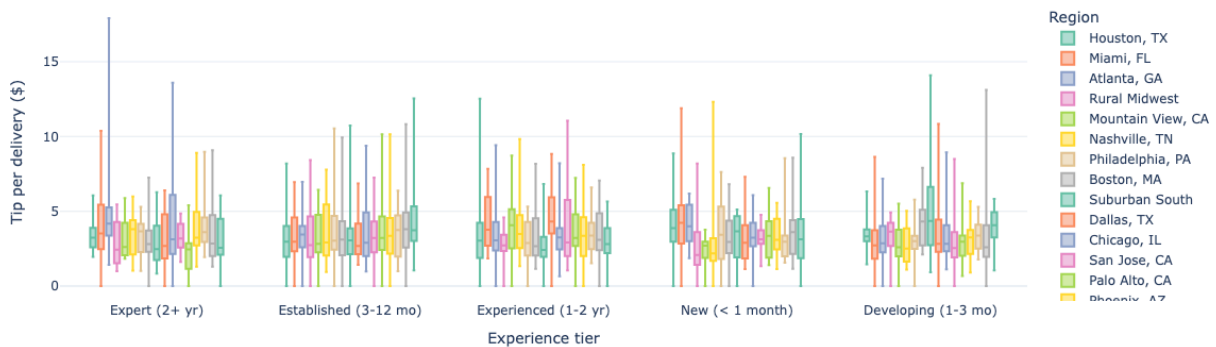
Earnings spread by region — P10 / median / P90



What this chart reveals that the box plot does not is how uniform the risk profile is across every market. Every region has its P10 starting around \$4–5/hr and its P90 reaching \$18–22/hr, and the median diamonds are all packed between \$9 and \$11. There is no clearly safe market with a meaningfully higher floor and no clearly volatile market with a dramatically wider spread. The implication for the ROI tool is that advising a driver to switch markets to reduce earnings risk would not be well supported by this data. What the chart actually argues is that within-market decisions around time slot, vehicle efficiency, and active ratio are likely more impactful than where you choose to drive, which is a nuanced and honest conclusion.

3.10 Tip Per Delivery by Experience Tier

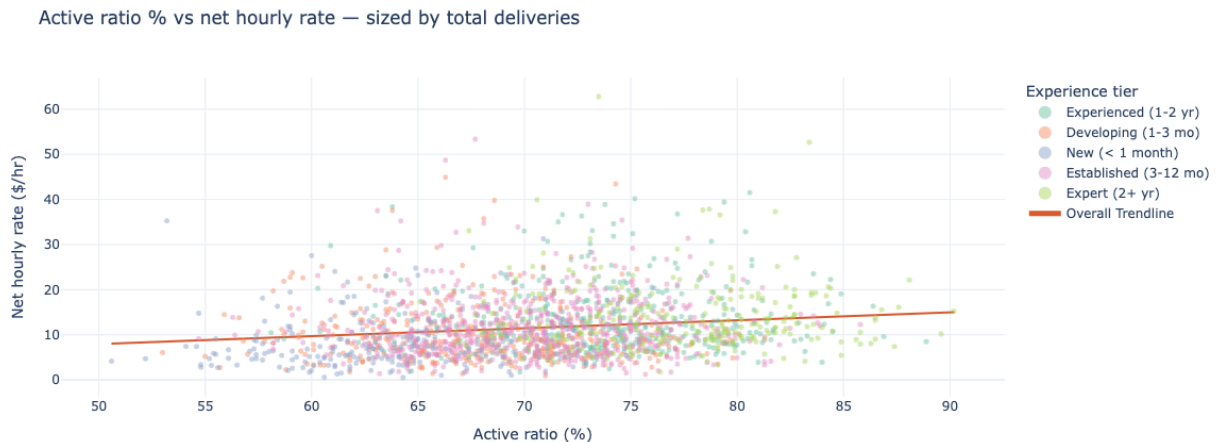
Tip per delivery by experience tier and region



The most honest observation from this chart is that it is visually noisy, and that noise is itself the finding. Expert drivers with two or more years of experience do not show a clearly higher median tip than drivers who started less than a month ago. The distributions overlap substantially across all five tiers. Houston and Suburban South tend to show slightly higher individual outliers across tiers, suggesting that regional tipping culture may matter more than how long someone has been driving. The practical takeaway is that tip income appears to be driven primarily by factors outside driver control, including order type, customer behavior, and platform defaults, rather than by accumulated experience. For the model, this means tips should be treated as a

noisy and region-influenced variable rather than something that improves predictably with tenure.

3.11 Active Ratio vs Net Hourly Rate



The trendline tells a clear and actionable story. As active ratio increases from 50% to 90%, net hourly rate rises from roughly \$8/hr to \$15/hr, nearly doubling effective pay through utilization alone without changing market, vehicle, or time slot. The scatter is wide, so a high active ratio does not guarantee strong earnings, but the direction is unambiguous and the slope is meaningful. Experience tier coloring does not show strong separation, with Expert and New drivers scattered equally throughout the cloud, which again suggests that how efficiently you use your time matters more than how long you have been doing the job. For the ROI tool, this is arguably the most actionable chart in the entire suite because active ratio is something a driver can partially control through smarter market timing and shift selection.

4 Conclusions and Final Statements

4.1 Key Findings

The exploratory analysis reveals several patterns across the dataset. Net hourly earnings are influenced by region, vehicle class, and time slot, but the results of those effects are not always intuitive. Medians across vehicle classes are uniform; the real differentiation occurs in the upper and lower tails, meaning vehicle choice affects earnings risk and upside more than it affects the average session. Fuel costs are the most consistent drag on net earnings for fuel-using vehicles, and that drag is measurably sensitive to gas price fluctuations. Drivers should weigh their decision heavily based on their vehicle's fuel efficiency. Active ratio, the fraction of online time spent on active deliveries, shows a positive association with net hourly rate, suggesting that time utilization should be one of the most valuable metrics for a driver to try and optimize.

4.2 Real-World Applications

The patterns identified in this analysis have direct practical applications for the driver ROI tool that this project is ultimately building toward. Specifically, the model can surface the following types of guidance:

- Market selection: regions with high medians and narrow quantile bands represent the most reliably profitable markets
- Shift timing: the region $\times$ time slot heatmap provides a data-backed starting point for recommending when to drive in a given market
- Vehicle assessment: drivers with EVs or hybrids can expect their fuel cost advantage to compound on longer, higher-earning sessions
- Efficiency benchmarking: a driver's active ratio can be compared against the distribution in the data to identify whether idle time is suppressing their hourly rate
- Gas price sensitivity: the model can simulate how a change in local gas prices would affect projected net earnings for a given vehicle class

4.3 Restatement of Major Assumptions

Before drawing any formal conclusions, all assumptions previously made must be acknowledged while interpreting any findings. Every finding in this report is conditional on these assumptions holding, and any deviation from them in a real-world deployment context could alter the results materially.

4.4 What We Cannot Officially Conclude

Given the synthetic nature of the data and the assumptions enumerated above, the following conclusions cannot be drawn from this analysis alone:

- We cannot conclude that any specific region, vehicle class, or time slot will produce the earnings ranges shown here for any actual driver on any actual platform.
- We cannot conclude that the relationships identified (between active ratio and earnings, gas price and net pay, or experience tier and tips) hold at a statistically validated level in the real world without validation with observed driver data.

- We cannot conclude that vehicle selection advice derived from this model would hold if platform pay algorithms change, as they frequently do.
- We cannot conclude that the correlation matrix reflects true causal relationships. Correlation identifies association, not causation, and in a synthesized dataset, some correlations may reflect the construction formula rather than genuine economic dynamics.
- We cannot conclude that the findings generalize to platforms other than DoorDash, or to international markets, without significant additional research.

4.5 Next Steps

The next phase of this project is statistical and machine learning modeling using the cleaned dataset. The correlation matrix and violin charts have produced a preliminary feature shortlist centered on `active_ratio_pct`, `vehicle_class`, `region`, `time_slot`, `avg_gas_price`, and `deliveries_per_hour`. A regularized linear regression will be fitted first as an interpretable baseline, followed by gradient boosting models that can capture nonlinear interactions between features. Quantile regression will be used to produce earnings ranges rather than point estimates, which is the output format the ROI tool ultimately requires. Validation will be conducted against published driver earnings benchmarks to assess whether the model's predictions are calibrated to the real-world range, even if they cannot be validated against a held-out sample of actual sessions.